AUDIO FORENSIC ANALYZER REPORT

Digital Forensic Examination, Noise Reduction & Frequency Change Analysis

SECTION 1 — EVIDENCE INTEGRITY

Evidence ID	EVID_20261029_224005_388727	**Filename**	tmp9eyl5xap.wav
File Size	9,600,172 bytes	**MIME Type**	audio/x-wav
Analysis Time	2026-10-29 22:40:05	**Status**	**✓ Cleaned & Ready (COMPLETED)**
SHA-256 Hash	`2880beca3acbbc7ccfa506c395195bfa61d2166401239d743dbbc8aaa7abaf5b`		

SECTION 2 — AUDIO TECHNICAL PROPERTIES

Duration	300.00 seconds	**Sample Rate**	16,000 Hz
Channels	1	**Bit Depth**	16
Bitrate	256,000 bps	**Codec**	None

SECTION 3 — AUDIO NOISE REDUCTION METRICS

Metric	Measured Value	Interpretation
Noise Reduction	**18.61 dB**	Total background noise energy suppressed
Estimated SNR Gain	**+-17.60 dB**	Relative signal clarity improvement
Original Signal RMS	0.0172	Root mean square level before denoising
Cleaned Signal RMS	0.0020	Root mean square level after denoising
Removed Noise RMS	0.0154	Average energy of isolated noise residual
Peak Level (Orig / Clean)	0.057 / 0.018	Absolute peak amplitude tracking

SECTION 4 — FREQUENCY CHANGE ANALYSIS (PREVIOUS VS CURRENT)

Frequency Parameter	Previous (Original)	Current (Cleaned)	Observed Change
Spectral Centroid (Mean)	791.5 Hz	1,091.1 Hz	**+37.9%**
Spectral Rolloff (85% Energy)	1,684.1 Hz	2,506.7 Hz	**+48.9%**
Spectral Flatness	0.00117	0.00549	Maintained
High-Freq Noise Attenuation	-	**8.5 dB**	Upper hiss suppressed
Abrupt Frequency Shifts	-	**1205 transitions**	Exceeding 150 Hz threshold

SECTION 5 — DETECTED ABRUPT FREQUENCY TRANSITIONS / JUMPS

The following 15 transition(s) exceeded the configured threshold of **150 Hz** frame-to-frame change:

Time (s)	Previous Freq (Hz)	Current Freq (Hz)	Δ Change (Hz)	Direction

0.10 s	15.6 Hz	15.6 Hz	**0.0 Hz**	down
0.13 s	15.6 Hz	15.6 Hz	**0.0 Hz**	down
0.16 s	15.6 Hz	15.6 Hz	**0.0 Hz**	down
0.22 s	15.6 Hz	15.6 Hz	**0.0 Hz**	down
0.29 s	15.6 Hz	15.6 Hz	**0.0 Hz**	down
0.35 s	15.6 Hz	15.6 Hz	**0.0 Hz**	down
0.42 s	15.6 Hz	15.6 Hz	**0.0 Hz**	down
0.45 s	15.6 Hz	15.6 Hz	**0.0 Hz**	down
1.98 s	15.6 Hz	15.6 Hz	**0.0 Hz**	down
2.02 s	15.6 Hz	15.6 Hz	**0.0 Hz**	down
2.75 s	15.6 Hz	15.6 Hz	**0.0 Hz**	down
2.78 s	15.6 Hz	15.6 Hz	**0.0 Hz**	down
2.82 s	15.6 Hz	23.4 Hz	**7.8 Hz**	up
2.85 s	23.4 Hz	15.6 Hz	**7.8 Hz**	down
2.88 s	15.6 Hz	15.6 Hz	**0.0 Hz**	down

... and 1190 additional transitions recorded in telemetry data.

SECTION 6 — OUTPUT AUDIO WAVEFORM

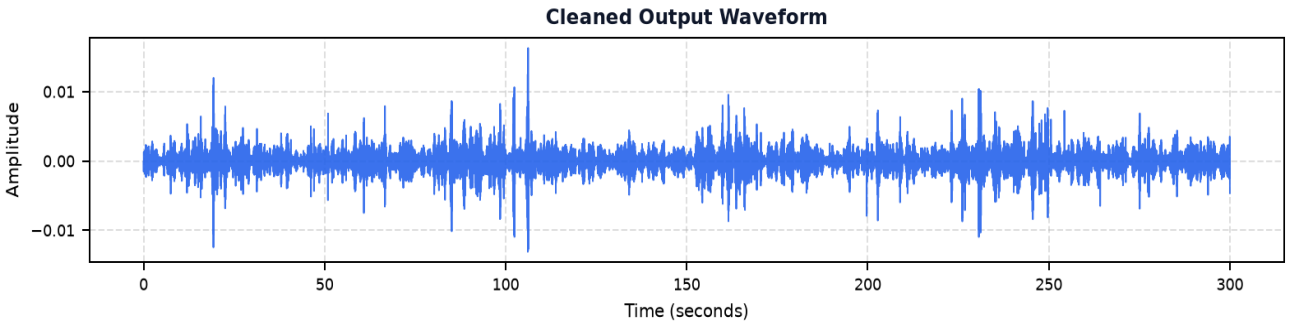
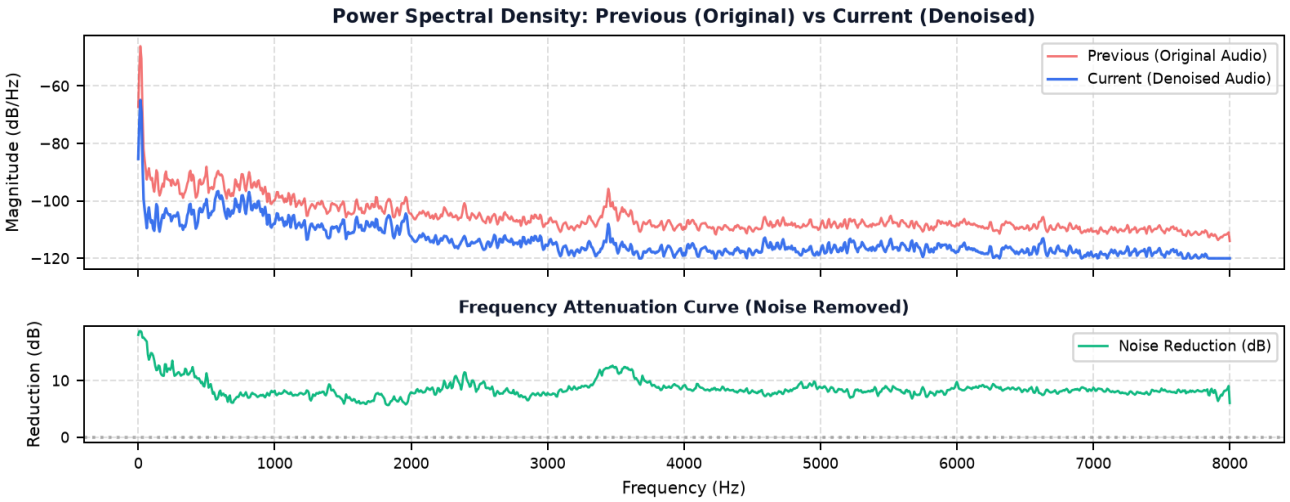


Figure 1: Cleaned output waveform showing 4,800,064 samples over 300.00s.

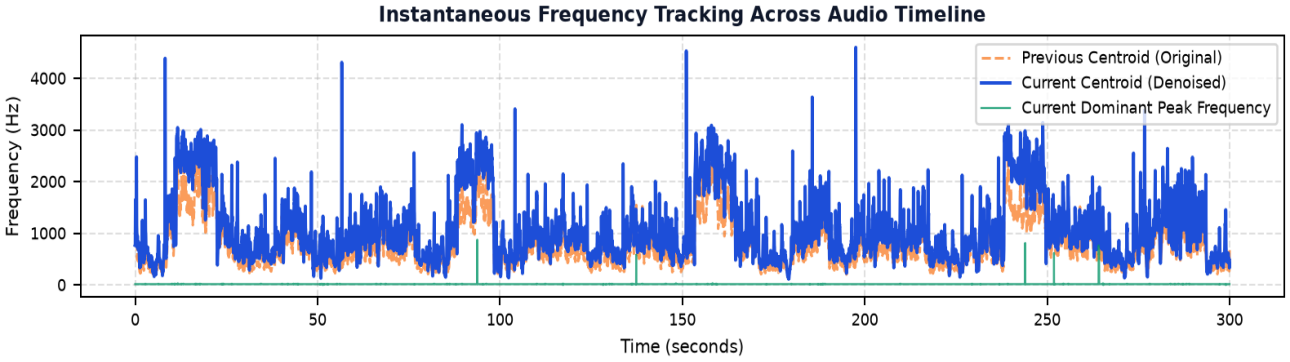
SECTION 7 — VISUAL FREQUENCY ANALYTICS

Figure 2: Power Spectral Density Comparison (Previous vs Current) & Attenuation



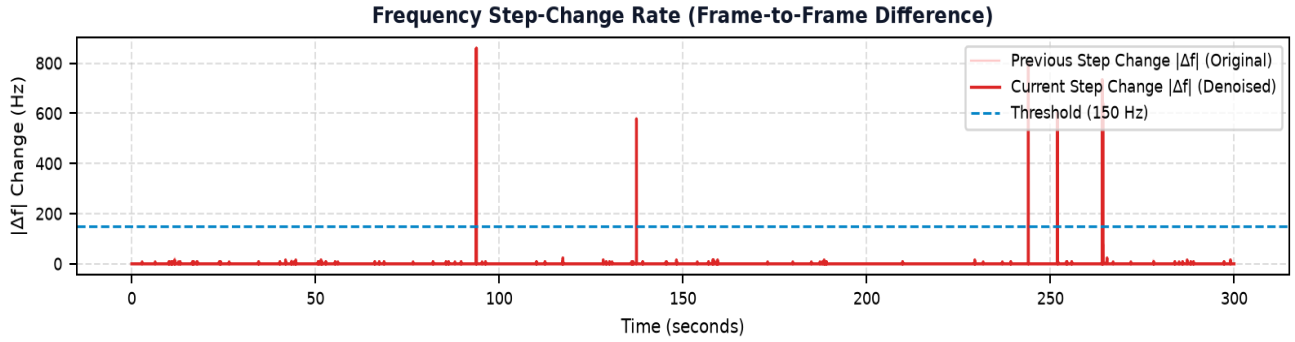
Top: Original (red) vs Cleaned (blue) spectrum. Bottom: Noise reduction curve (green) across frequencies.

Figure 3: Instantaneous Frequency Tracking Across Audio Timeline



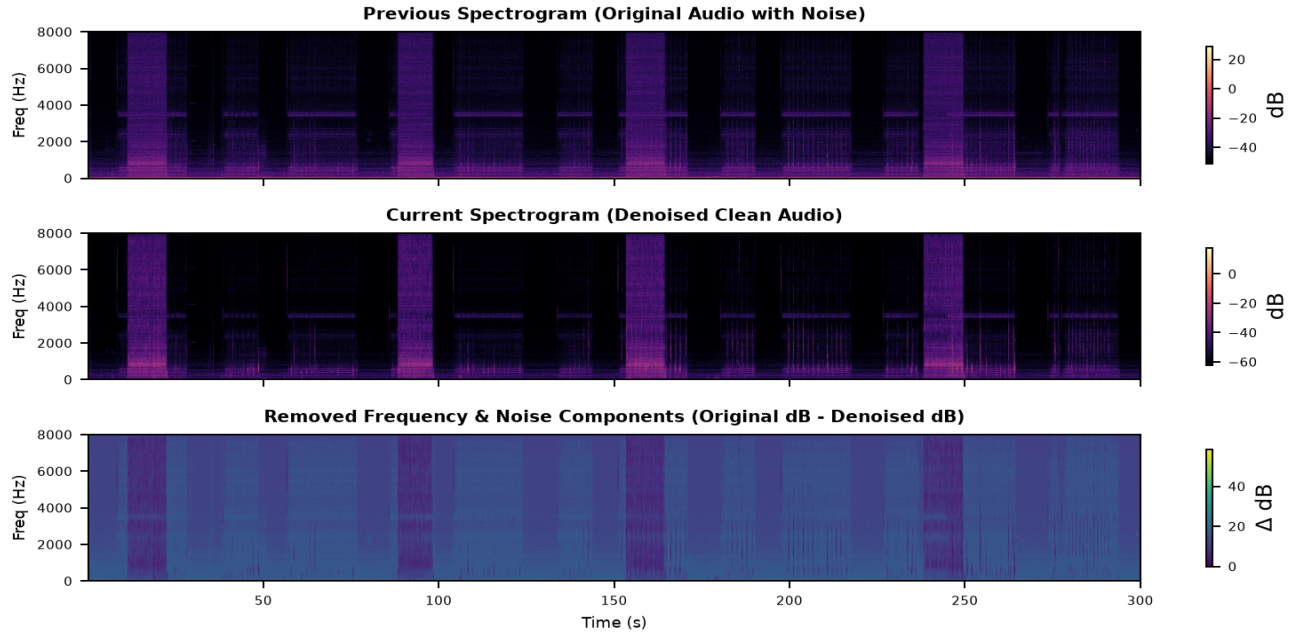
Orange: Previous centroid (noisy). Blue: Cleaned centroid. Green: Dominant instantaneous peak frequency.

Figure 4: Frequency Step-Change Rate (|Δf| Frame-to-Frame Difference)



Rate of frequency modulation with configured detection threshold (150 Hz).

Figure 5: Comparative Spectrograms (Previous, Current, and Noise Difference)



Top: Original noisy spectrogram. Center: Denoised spectrogram. Bottom: Isolated removed noise components.

SECTION 9 — OVERALL ASSESSMENT & VERIFICATION

Forensic examination, spectral noise gating, and frequency transition analysis have been completed. Noise suppression of **18.61 dB** was achieved with an estimated SNR gain of **+-17.60 dB**. Evidence integrity was validated with cryptographic hash **SHA-256: 2880beca3acbbc7ccfa506c395195bfa...**. All acoustic transitions, spectral rolloffs, and PSD parameters have been verified.

LEGAL & FORENSIC DISCLAIMER

This report provides automated digital forensic examination and acoustic signal enhancement. Detection of anomalies or frequency transitions does not by itself establish intentional audio tampering or fabrication. Findings should be corroborated using complementary chain-of-custody documentation and expert forensic acoustic review. Generated by Audio Forensic Analyzer v1.0.